

Measures of Central Tendency

Introduction:

According to Prof Bowley “Measures of central tendency (averages) are statistical constants which enable us to comprehend in a single effort the significance of the whole.” Measure of central tendency is a measure that tells us where the middle of a bunch of data lies. One of the most widely used set of summary figures is known as measures of location which are often referred to as averages, central tendency or central location. An average is a single value which is considered as the most representative or typical value for a given set of data. Such a value is neither the smallest nor the larger value, but is a number whose value is somewhere in the middle of the group. For this reason an average is frequently referred to as a measure of central tendency or central value.

Objectives of Averaging:

The main objectives of Measure of Central Tendency are

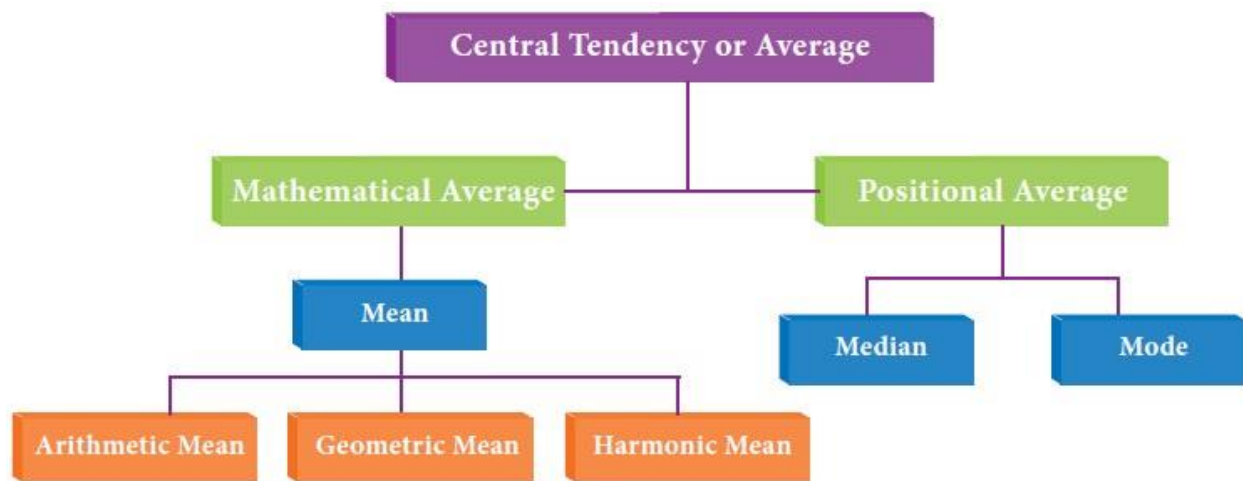
- 1) To get one single value that describes the characteristic of the entire data.
- 2) To facilitate comparisons between data.

Characteristics for a good or an ideal average:

There are different types of averages, each has its own advantages and disadvantages. Requisites of a Good Measure of Central Tendency:

1. It should be rigidly defined.
2. It should be easy to understand and compute.
3. It should be based on all items in the data.
4. Its definition shall be in the form of a mathematical formula.
5. It should be capable of further algebraic treatment.
6. It should have sampling stability.
7. It should be capable of being used in further statistical computations or processing.

Various measures of central tendency



Others partition values are Quartiles, Deciles and Percentiles.

Arithmetic Mean

Definition of Arithmetic Mean (Average):

Arithmetic mean is commonly known as average. The average of a given set of numbers is called the arithmetic mean, or simply, the mean of the given numbers.

Thus, the arithmetic mean of a group of observations is defined as ...

Mean = (Sum of observations)/(Number of observations)

To find A.M. for Raw data : For a raw data, the arithmetic mean of a series of numbers is sum of all observations divided by the number of observations in the series. Thus, the mean of n observation $x_1, x_2, \dots, x_n$, is given by A.M

$$\bar{X} = \frac{(x_1 + x_2 + \dots + x_n)}{n} = \frac{\sum_{i=1}^n x_i}{n},$$

Where the symbol $\sum$ called *sigma*, stands for summation, and we write,

$$(x_1 + x_2 + \dots + x_n) = \sum_{i=1}^n x_i$$

Then arithmetic mean (A.M.) for n observations is: (direct method)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

There are two methods for computing the A.M :

(i) Direct method

(ii) Short cut method.

Example The following data represent the number of books issued in a school library on selected from 7 different days 7, 9, 12, 15, 5, 4, 11 find the mean number of books.

Solution:

Arithmetic mean

$$\begin{aligned}\bar{x} &= \frac{1}{n} \sum_{i=1}^n x_i \\ \bar{x} &= \frac{7+9+12+15+5+4+11}{7} \\ &= \frac{63}{7} = 9\end{aligned}$$

Hence the mean of the number of books is 9

Example

From the following data, calculate the arithmetic mean.

23 25 26 27 30 31 32 33 34 34 36

Solution:

$$\text{Arithmetic Mean } (\bar{X}) = \frac{\Sigma X}{n} = \frac{23+25+26+27+30+31+32+33+34+34+36}{11} = \frac{331}{11} = 30.09$$

[Ans.]

A.M.

Under this method an assumed mean or an arbitrary value (denoted by A) is used as the basis of calculation of deviations (d_i) from individual values. That is if $d_i = x_i - A$

Then

$$\bar{x} = A + \frac{\sum_{i=1}^n d_i}{n}$$

Example

A student's marks in 5 subjects are 75, 68, 80, 92, 56. Find the average of his marks.

Solution:

Let us take the assumed mean, $A = 68$

x_i	$d_i = x_i - 68$
75	7
68	0
80	12
56	-12
92	24
Total	31

$$\begin{aligned}
 \bar{x} &= A + \frac{\sum_{i=1}^n d_i}{n} \\
 &= 68 + \frac{31}{5} \\
 &= 68 + 6.2 = 74.2
 \end{aligned}$$

The arithmetic mean of average marks is 74.2

(b) To find A.M. for Discrete Grouped data

If $x_1, x_2, \dots, x_n$ are discrete values with the corresponding frequencies $f_1, f_2, \dots, f_n$. Then the mean for discrete grouped data is defined as (direct method)

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

In the short cut method the formula is modified as

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i d_i}{N} \quad \text{where } d_i = x_i - A$$

Example A proof reads through 73 pages manuscript The number of mistakes found on each of the pages are summarized in the table below Determine the mean number of mistakes found per page

No of mistakes	1	2	3	4	5	6	7
No of pages	5	9	12	17	14	10	6

Solution:

(i) Direct Method

x_i	f_i	$f_i x_i$
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
Total	N=73	299

$$\begin{aligned}
 \bar{x} &= \frac{\sum_{i=1}^n f_i x_i}{N} \\
 &= \frac{299}{73} \\
 &= 4.09
 \end{aligned}$$

The mean number of mistakes is 4.09

(ii) Short-cut Method

x_i	f_i	$d_i = x_i - A$	$f_i d_i$
1	5	-3	-15
2	9	-2	-18
3	12	-1	-12
4	17	0	0
5	14	1	14
6	10	2	20
7	6	3	18
	$\Sigma f_i = 73$		$\Sigma f_i d_i = 7$

$$\begin{aligned}\bar{x} &= A + \frac{\sum_{i=1}^n f_i d_i}{N} \\ &= 4 + \frac{7}{73} \\ &= 4.09\end{aligned}$$

The mean number of mistakes = 4.09

(c) Mean for Continuous Grouped data:

For the computation of A.M for the continuous grouped data, we can use direct method or short cut method.

Direct Method:

The formula is

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N}, \quad x_i \text{ is the midpoint of the class interval}$$

Short cut method

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i d_i}{N} \times C$$

$$d = \frac{x_i - A}{c}$$

where A - any arbitrary value
 c - width of the class interval

x_i is the midpoint of the class interval.

Example The following the distribution of persons according to different income groups

Income (in ` 1000)	0 – 8	8 – 16	16 – 24	24 – 32	32 – 40	40 – 48
No of persons	8	7	16	24	15	7

Find the average income of the persons.

Solution :

Direct Method:

<i>Class</i>	f_i	x_i	$f_i x_i$
0-8	8	4	32
8 – 16	7	12	84
16-24	16	20	320
24-32	24	28	672
32-40	15	36	540
40-48	7	44	308
Total	N =77		1956

$$\begin{aligned}\bar{x} &= \frac{\sum_{i=1}^n f_i x_i}{N} \\ &= \frac{1956}{77} \\ &= 25.40\end{aligned}$$

N= $\sum f_i$

Example: The following table gives the daily wages distribution of the employees of a commercial organization:

Daily wages (Tk)	No. of employees
250 – 300	25
300 – 350	30
350 – 400	55
450 – 500	67
500 – 550	39
550 – 600	14

Compute arithmetic mean.

Solution:

Table for calculation:

Daily wages(Class interval)	Midpoint (xi)	No.of employees(fi)	fi xi
250 – 300	275	25	6875
300 – 350	325	30	9750
350 – 400	375	55	20625
400 – 450	425	0	0
450 – 500	475	67	31825
500 – 550	525	39	20475
550 – 600	575	14	8050
		N = 230	$\sum fixi = 97600$

$$\text{Arithmetic mean } (\bar{X}) = \frac{\sum fixi}{N} = \frac{97600}{230} = 424.35 \quad [\text{Ans.}]$$

Short cut method:

Class	f_i	x_i	$d_i = (x_i - A)/c$	$f_i d_i$
0 – 8	8	4	-3	-24
8 – 16	7	12	-2	-14
16 – 24	16	20	-1	-16
24 – 32	24	28 A	0	0
32 – 40	15	36	1	15
40 – 48	7	44	2	14
Total	N= 77			-25

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i d_i}{N} \times C$$

$$= 28 + \frac{-25}{77} \times 8 = 25.40$$

Example: Calculate the arithmetic mean by short cut from the following table:

Wages (Tk)	5-15	15-25	25-35	35-45	45-55
No.of workers	4	5	3	2	5

Solution:

Wages	No.of workers(f_i)	x_i	$d_i = (x_i - A)/C$	$f_i d_i$
5-15	4	10	-2 (10-30)/10	-8
15-25	5	20	-1	-5
25-35	3	30	0	0
35-45	2	40	1	2
45-55	5	50	2	10
				$\sum f_i d_i = -1$

Where $A=30$, $C=10$

Arithmetic mean

$$\bar{x} = A + \frac{\sum f_i d_i}{N} \times C$$

$$\bar{x} = 30 + \frac{-1}{19} \times 10$$

$$=30-0.53=29.47(\text{Ans})$$

Properties Arithmetic mean

Property 1 :

The algebraic sum of deviations of a set of observations from their arithmetic mean is zero.

That is,

for unclassified data, $\sum(x - \bar{x}) = 0$.

And for a grouped frequency distribution, $\sum f(x - \bar{x}) = 0$.

For example, if a variable "x" assumes five observations, say 10, 20, 30, 40, 50, then $\bar{x} = 30$.

The deviations of the observations from arithmetic mean $(x - \bar{x})$ are

$$=(10-30) + (20-30) + (30-30) + (40-30) + (50-30)$$

$$\text{Now, } \sum(x - \bar{x}) = (-20) + (-10) + 0 + 10 + 20 = 0$$

Since the sum of the deviations in the positive direction is equal to the sum of the deviations in the negative direction, the arithmetic mean is regarded as a measure of central tendency.

Property 2:

The sum of the squared deviations of the individual items from the arithmetic mean is always minimum. In other words, the sum of the squared deviations taken from any value other than the arithmetic mean will be higher.

Property 3:

As the arithmetic mean is based on all the items in a series, a change in the value of any item will lead to a change in the value of the arithmetic mean.

Property 4 :

Combined Mean:

Let $\bar{x}_1$ and $\bar{x}_2$ are the arithmetic mean of two groups (having the same unit of measurement of a variable), based on n_1 and n_2 observations respectively. Then the combined mean can be calculated using

$$\text{Combined Mean} = \bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Remark : The above result can be extended to any number of groups.

Example

A class consists of 4 boys and 3 girls. The average marks obtained by the boys and girls are 20 and 30 respectively. Find the class average.

Solution:

$$n_1 = 4, \bar{x}_1 = 20, n_2 = 3, \bar{x}_2 = 30$$

$$\begin{aligned} \text{Combined Mean} = \bar{x}_{12} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \\ &= \left[\frac{4 \times 20 + 3 \times 30}{4 + 3} \right] \\ &= \left[\frac{80 + 90}{7} = \frac{170}{7} \right] = 24.3 \end{aligned}$$

To Calculate Combined $\bar{X}$ for two or more series

For combined means ($\bar{X}_1, \bar{X}_2, \dots, \bar{X}_n$), we use the formula :-

$$\bar{X}_{123\dots n} = \frac{N_1\bar{X}_1 + N_2\bar{X}_2 + N_3\bar{X}_3 + \dots + N_n\bar{X}_n}{N_1 + N_2 + N_3 + \dots + N_n}$$

Note :- In case $N_1 = N_2 = N_3 = \dots = N_n$; Then a simple formula can be used.

$$\bar{X} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \dots + \bar{X}_n}{n} \quad \dots [n \text{ denotes the number of series}]$$

Example 1. Find combined $\bar{X}$ from following data :

	Firm I	Firm II	Firm III
Mean Wage (Rs.) :	12	15	18
No. of Workers :	20	40	30

Solution

Given : $\bar{X}_1 = \text{Rs. } 12$, $\bar{X}_2 = \text{Rs. } 15$, $\bar{X}_3 = \text{Rs. } 18$ And $N_1 = 20$, $N_2 = 40$, $N_3 = 30$.

$$\begin{aligned} \therefore \bar{X} &= \frac{N_1\bar{X}_1 + N_2\bar{X}_2 + N_3\bar{X}_3}{N_1 + N_2 + N_3} = \frac{12 \times 20 + 15 \times 40 + 18 \times 30}{20 + 40 + 30} \\ &= \frac{240 + 600 + 540}{90} = \frac{1380}{90} = \text{Rs. } 15.33. \text{ (approx.)} \end{aligned}$$

Problem

The mean of 100 workers in a factory running in two shifts of 60 and 40 workers respectively is Tk 38. The mean wage of 60 workers working in the morning shift is Tk 40. Find out the mean wage of the worker in evening shift.

Solution:

$$\begin{aligned} \text{Combined Mean : } \bar{X}_{12} &= \frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1 + N_2} \\ \Rightarrow 38 &= \frac{60 \times 40 + 40 \times \bar{X}_2}{60 + 40} \\ \Rightarrow 38 \times 100 &= 2400 + 40 \times \bar{X}_2 \\ \Rightarrow 3800 - 2400 &= 40 \times \bar{X}_2 \\ \Rightarrow 40 \times \bar{X}_2 &= 1400 \\ \Rightarrow \bar{X}_2 &= 35 \text{ [Ans.]} \end{aligned}$$

Merits

- ☐ It is easy to compute and has a unique value.
- ☐ It is based on all the observations.

- ☐ It is well defined. $55 \ 8 \ 2 \ 5+8+(-6)+0+2/5$
- ☐ It is least affected by sampling fluctuations. $10-15 \ 15-20 \ 20+$
- ☐ It can be used for further statistical analysis.

Limitations

- ☐ The mean is unduly affected by the extreme items (outliers).
- ☐ It cannot be determined for the qualitative data such as beauty, honesty etc.
- ☐ It cannot be located by observations on the graphic method.
- ☐ It cannot be calculated for open-end classes

When to use?

Arithmetic mean is a best representative of the data if the data set is homogeneous. On the other hand if the data set is heterogeneous the result may be misleading and may not represent the data.

Weighted Arithmetic Mean

The arithmetic mean, as discussed earlier, gives equal importance (or weights) to each observation in the data set. However, there are situations in which values of individual observations in the data set are not of equal importance. Under these circumstances, we may attach, a weight, as an indicator of their importance to each observation value.

Definition

Let $x_1, x_2, \dots, x_n$ be the set of n values having weights $w_1, w_2, \dots, w_n$ respectively, then the weighted mean is,

$$\bar{x}_w = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Uses of weighted arithmetic mean

Weighted arithmetic mean is used in:

- ☐ The construction of index numbers.
- ☐ Comparison of results of two or more groups where number of items in the groups differs.
- ☐ Computation of standardized death and birth rates.

Example 5.5

The weights assigned to different components in an examination or Component Weightage Marks scored

Component	Weightage	Marks scored
Theory	4	60
Practical	3	80
Assignment	1	90
Project	2	75
	10	

Calculate the weighted average score of the student who scored marks as given in the table

Solution:

Component	Marks scored (x_i)	Weightage (w_i)	$w_i x_i$
Theory	60	4	240
Practical	80	3	240
Assignment	90	1	90
Project	75	2	150
Total		10	720

Weighted average, $\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$

$$= 720/10$$

$$= 72$$

Symbilically ; $\bar{X}_w = \frac{\sum WX}{\sum W}$

Where X is variable, W is assigned weight

and $\bar{X}_w$ is weighted Arithmetic Mean.

Very Important Note :

To achieve $\bar{X}_w$ we use the same procedure as is used to find $\bar{X}$. In weighted A.M., W is taken instead of f. Formula is $\frac{\sum WX}{\sum W}$ instead of $\frac{\sum fX}{\sum f}$.

Example 1. Following are given the marks obtained and weights assigned to the subjects of a student. Calculate Weighted A.M.

Subject	English	Punjabi	Economics	Mathematics	Accounts
Marks :	60	65	53	50	40
Weight :	1	1	2	3	4

Solution

Subject	Marks X	Weights (W)	Product (XW)
English	60	1	60
Punjabi	65	1	65
Economics	53	2	106
Mathematics	50	3	150
Accounts	40	4	160
		$\sum W = 11$	$\sum XW = 541$

$$\begin{aligned} \sum XW &= 541 ; \sum W = 11 \\ \therefore \bar{X}_w &= \frac{\sum XW}{\sum W} = \frac{541}{11} \\ &= 49.18. \end{aligned}$$

Problems (missing frequencies)

From the following data find out the missing frequency when mean is 22.

Central Value 10 15 20 25 30

Frequency 5 10 15 ? 5

Solution:

Calculation of Original Frequency Distribution and Mean

Central Value (X)	f	fX
10	5	50
15	10	150
20	15	300
25	f_1	$25f_1$
30	5	150
	$N = 35 + f_1$	$\Sigma fX = 650 + 25f_1$

$$\therefore \text{Mean} = \frac{\Sigma fX}{N}$$

$$\Rightarrow 22 = \frac{650 + 25f_1}{35 + f_1}$$

$$\Rightarrow 770 + 22 f_1 = 650 + 25f_1$$

$$\Rightarrow 3f_1 = 120$$

$$\Rightarrow f_1 = 40$$

$\therefore$ The required missing frequency (f_1) = 40 [Ans.]

The mean salary paid to 200 employees in a Garment factory was found to be Tk 2500. Later on, it was discovered that the salaries of two employees were wrongly taken as 3000 and 3500 instead of 3500 and 3200. Compute the correct mean salary.

Solution:

Given $N = 200$

$$\bar{X} = 2500$$

We know, $\bar{X} = \frac{\Sigma X}{N}$

$$\Rightarrow 2500 = \frac{\Sigma X}{200}$$

$$\Rightarrow \Sigma X = 2500 \times 200$$

$$\Rightarrow \Sigma X = 500000, \text{ but this is not correct.}$$

$$\begin{aligned} \therefore \text{Correct } \Sigma X &= 500000 - 3000 - 3500 + 3500 + 3200 \\ &= 500200 \end{aligned}$$

$$\therefore \text{Correct Mean } (\bar{X}) = \frac{\text{Correct } \Sigma X}{N} = \frac{500200}{200} = 2501 \quad [\text{Ans.}]$$

The mean of 200 was 50. Later on it was found that two observations were wrongly read as 92 and 8 instead of 192 and 88 respectively. Find the correct mean.

Solution: We know, $\bar{X} = \Sigma x/N$

$$\Rightarrow 50 = \Sigma x/200$$

$$\Rightarrow \Sigma x = 10000, \text{ which is wrong.}$$

$$\text{So correct } \Sigma x = 10000 - 92 - 8 + 192 + 88 = 10180$$

$$\text{So correct } \bar{X} = \Sigma x/N = 10180/200 = 50.90 \quad [\text{Ans.}]$$

If the mean of 9, 8, 10, x, 12 is 15, find the value of x.

Solution:

$$\text{Mean of the given numbers} = (9 + 8 + 10 + x + 12)/5 = (39 + x)/5$$

According to the problem, mean = 15 (given).

$$\text{Therefore, } (39 + x)/5 = 15$$

$$\Rightarrow 39 + x = 15 \times 5$$

$$\Rightarrow 39 + x = 75$$

$$\Rightarrow 39 - 39 + x = 75 - 39$$

$$\Rightarrow x = 36$$

Hence, $x = 36$.

If the mean of five observations x, x + 4, x + 6, x + 8 and x + 12 is 16, find the value of x.

Solution: Mean of the given observations

$$= x + (x + 4) + (x + 6) + (x + 8) + (x + 12)/5$$

$$= (5x + 30)/5$$

According to the problem, mean = 16 (given).

$$\text{Therefore, } (5x + 30)/5 = 16$$

$$\Rightarrow 5x + 30 = 16 \times 5$$

$$\Rightarrow 5x + 30 = 80$$

$$\Rightarrow 5x + 30 - 30 = 80 - 30$$

$$\Rightarrow 5x = 50$$

$$\Rightarrow x = 50/5$$

$$\Rightarrow x = 10$$

Hence, $x = 10$.

$$148 + 153 + 146 + 147 + 154$$

The mean of 40 numbers was found to be 38. Later on, it was detected that a number 56 was misread as 36. Find the correct mean of given numbers.

Solution:

Calculated mean of 40 numbers = 38.

Therefore, calculated sum of these numbers = $(38 \times 40) = 1520$.

Correct sum of these numbers

$$= [1520 - (\text{wrong item}) + (\text{correct item})]$$

$$= (1520 - 36 + 56)$$

$$= 1540.$$

Therefore, the correct mean = $1540/40 = 38.5$.

The mean of the heights of 6 boys is 152 cm. If the individual heights of five of them are 151 cm, 153 cm, 155 cm, 149 cm and 154 cm, find the height of the sixth boy.

Solution:

Mean height of 6 boys = 152 cm.

Sum of the heights of 6 boys = $(152 \times 6) = 912$ cm

Sum of the heights of 5 boys = $(151 + 153 + 155 + 149 + 154)$ cm = 762 cm.

Height of the sixth boy

= (sum of the heights of 6 boys) - (sum of the heights of 5 boys)

= (912 - 762) cm = 150 cm.

Hence, the height of the sixth girl is 150 cm.

Example: We will discuss here how to find the sum of first n natural numbers.

Let S be the required sum.

Therefore, $S = 1 + 2 + 3 + 4 + 5 + \dots + n$

Clearly, it is an Arithmetic Progression whose first term = 1, last term = n and number of terms = n .

Therefore, $S = \frac{n}{2}(n + 1)$, [Using the formula $S = \frac{n}{2}(a + l)$]

Solved examples to find the sum of first n natural numbers

1. Find the sum and mean of first 25 natural numbers.

Solution:

Let S be the required sum.

Therefore, $S = 1 + 2 + 3 + 4 + 5 + \dots + 25$

Clearly, it is an Arithmetic Progression whose first term = 1, last term = 25 and number of terms = 25.

Therefore, $S = \frac{25}{2}(25 + 1)$, [Using the formula $S = \frac{n}{2}(a + l)$]

$$= \frac{(25 \times 26)}{2}$$

$$= 25 \times 13$$

$$= 325$$

$$\text{Mean} = \frac{325}{25} = 13$$

Therefore, the sum and mean of first 25 natural numbers are 325 and 13.

Geometric Mean (GM)

(a) G.M. For Ungrouped data

The Geometric Mean (G.M.) of a set of n observations is the n th root of their product. If $x_1, x_2, \dots, x_n$ are n observations (observations are positive and non zero) then

$$\text{G. M.} = \sqrt[n]{x_1 x_2 \dots x_n} = (x_1 x_2 \dots x_n)^{\frac{1}{n}}$$

Taking the n th root of a number is difficult. Thus, the computation is done as under

$$\text{Log G.M.} = \frac{1}{n} \log (x_1 x_2 \dots x_n)$$

$$= \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n)$$

$$= \frac{1}{n} \sum \log x_i$$

$$\text{G.M} = \text{Anti log} \left[\frac{1}{n} \sum \log x_i \right]$$

Solved Examples

Question 1: Find the geometric mean of 4 and 3 ?

Solution:

$$\text{Geometric Mean} = \sqrt{4 \times 3} = 2\sqrt{3} = 3.46$$

Question 2: Find the geometric mean of 9, 27 and 3 ?

Solution:

$$\text{Geometric Mean} = \sqrt[3]{9 \times 27 \times 3} = \sqrt[3]{729} = 9$$

3. Calculate the geometric mean of the following price relatives:

Commodity	Price relatives	Commodity	Price relatives
Wheat	237	Sugar	124
Rice	198	Salt	107
Pulses	156	Oils	196

Solution: We know,

$$\begin{aligned}\text{G.M.} &= (X_1 \cdot X_2 \cdot X_3 \cdot X_4 \cdot X_5 \cdot X_6)^{\frac{1}{n}} \\ &= (237 \times 198 \times 156 \times 124 \times 107 \times 196)^{\frac{1}{6}} \\ &= (19037050800768)^{0.1667} \\ &= 163.57 \quad [\text{Ans}]\end{aligned}$$

Example : Calculate the geometric mean of the annual percentage growth rate of profits in business corporate from the year 2000 to 2005 is given below

50, 72, 54, 82, 93

Solution:

x_i	50	72	54	82	93	Total
$\log x_i$	1.6990	1.8573	1.7324	1.9138	1.9685	9.1710

$$\begin{aligned}
 \text{G.M.} &= \text{Antilog} \frac{\sum_{i=1}^n \log x_i}{n} \\
 &= \text{Antilog} \frac{9.1710}{5} \\
 &= \text{Antilog } 1.8342
 \end{aligned}$$

$$\text{G. M.} = 68.26$$

Geometrical mean of annual percentage growth rate of profits is 68.26

Example

The population in a city increased at the rate of 15% and 25% for two successive years. In the next year it decreased at the rate of 5%. Find the average rate of growth.

Solution:

Let us assume that the population is 100

Percentage rise in population	Population at the end of year x_i	$\log x_i$
15	115	2.0607
25	125	2.0969
5	95	1.9777
		6.1353

$$\begin{aligned}
 \text{G.M} &= \text{Antilog } \frac{\sum_{i=1}^n \log x_i}{n} \\
 &= \text{Antilog } \frac{(6.1353)}{3} \\
 &= \text{Antilog } (2.0451) \\
 &= 110.9
 \end{aligned}$$

(b) G.M. For Discrete grouped data

If $x_1, x_2, \dots, x_n$ are discrete values of the variate x with corresponding frequencies $f_1, f_2, \dots, f_n$. Then geometric mean is defined as

$$\text{G. M.} = \text{Antilog } \frac{\sum_{i=1}^n f_i \log x_i}{N} \text{ with usual notations}$$

Example : Find the G.M for the following data, which gives the defective screws obtained in a factory.

Diameter (cm)	5	15	25	35
Number of defective screws	5	8	3	4

Solution:

x_i	f_i	$\log x_i$	$f_i \log x_i$
5	5	0.6990	3.4950
15	8	1.1761	9.4088
25	3	1.3979	4.1937
35	4	1.5441	6.1764
	N=20		23.2739

$$\begin{aligned}
 \text{G.M} &= \text{Antilog} \frac{\sum_{i=1}^n f_i \log x_i}{N} \\
 &= \text{Antilog} \frac{23.2739}{20} \\
 &= \text{Antilog } 1.1637
 \end{aligned}$$

$$\text{G.M} = 14.58$$

(c) G.M. for Continuous grouped data

Let x_i be the mid point of the class interval

$$\text{G. M.} = \text{Antilog} \left[\frac{\sum_{i=1}^n f_i \log x_i}{N} \right]$$

Example : The following is the distribution of marks obtained by 109 students in a subject in an institution. Find the Geometric mean.

Marks	4-8	8-12	12-16	16-20	20-24	24-28	28-32	32-36	36-40
No. of Students	6	10	18	30	15	12	10	6	2

Solution:

Marks	Mid point (x_i)	f_i	$\log x_i$	$f_i \log x_i$
4-8	6	6	0.7782	4.6692
8-12	10	10	1.0000	10.0000
12-16	14	18	1.1461	20.6298
16-20	18	30	1.2553	37.6590
20-24	22	15	1.3424	20.1360
24-28	26	12	1.4150	16.8000
28-32	30	10	1.4771	14.7710
32-36	34	6	1.5315	9.1890
36-40	38	2	1.5798	3.1596
Total		N = 109		137.1936

$$\begin{aligned}
 \text{G.M.} &= \text{Antilog} \left[\frac{\sum_{i=1}^n f_i \log x_i}{N} \right] \\
 &= \text{Antilog} \left[\frac{137.1936}{109} \right] = \text{Antilog} [1.2587]
 \end{aligned}$$

$$\text{G. M.} = 18.14$$

Geometric mean marks of 109 students in a subject is 18.14

Merits of Geometric Mean:

- ☐ It is based on all the observations
- ☐ It is rigidly defined
- ☐ It is capable of further algebraic treatment
- ☐ It is less affected by the extreme values
- ☐ It is suitable for averaging ratios, percentages and rates.

Limitations of Geometric Mean:

- ☐ It is difficult to understand
- ☐ The geometric mean cannot be computed if any item in the series is negative or zero.
- ☐ The GM may not be the actual value of the series
- ☐ It brings out the property of the ratio of the change and not the absolute difference of change as the case in arithmetic mean.

Harmonic Mean (H.M.)

Harmonic Mean is defined as the reciprocal of the arithmetic mean of reciprocals of the observations.

(a) H.M. for Ungrouped data

For ungrouped data: If $x_1, x_2, \dots, x_n$ are n observations (positive and nonzero), then Harmonic Mean, $H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$

$$H. M. = \frac{n}{\sum_{i=1}^n \left(\frac{1}{x_i} \right)}$$

i) 7 5 9 ii) 7 0 4 iii) 1 6 -3

$$H.M = \frac{3}{\frac{1}{7} + \frac{1}{5} + \frac{1}{9}}$$

$$H.M = \frac{3}{\frac{1}{7} + \frac{1}{0} + \frac{1}{4}}$$

$$H.M = \frac{3}{\frac{1}{1} + \frac{1}{6} + \frac{1}{-3}}$$

Example: Calculated H.M of data 2,7,12.

$$H.M = \frac{3}{\frac{1}{2} + \frac{1}{7} + \frac{1}{12}}$$

$$H.M = \frac{3}{0.5 + 0.142 + 0.083}$$

$$H.M = \frac{3}{0.725}$$

H.M=4.137

Example : A man travels from Dhaka to Jamalpur by a car and takes 4 hours to cover the whole distance. In the first hour he travels at a speed of 50 km/hr, in the second hour his speed is 64 km/hr, in third hour his speed is 80 km/hr and in the fourth hour he travels at the speed of 55 km/hr. Find the average speed of the motorist.

• **Solution:**

x	50	65	80	55	Total
$1/x$	0.0200	0.0154	0.0125	0.0182	0.0661

$$\begin{aligned}
 \text{H. M.} &= \frac{n}{\sum \left(\frac{1}{x_i} \right)} \\
 &= \frac{4}{0.0661} = 60.5 \text{ km/hr}
 \end{aligned}$$

Average speed of the motorist is 60.5km/hr

(b) H.M. for Discrete Grouped data:

For a frequency distribution

$$\text{Harmonic Mean, } H.M = \frac{N}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}} = \frac{N}{\sum \frac{f_i}{x_i}}$$

$$H. M. = \frac{N}{\sum_{i=1}^n f_i \left(\frac{1}{x_i} \right)}$$

Example The following data is obtained from the survey. Compute H.M

Speed of the car	130	135	140	145	150
No of cars	3	4	8	9	2

Solution:

x_i	f_i	$\frac{f_i}{x_i}$
130	3	0.0231
135	4	0.0091
140	8	0.0571
145	9	0.0621
150	2	0.0133
Total	N = 26	0.1648

$$\begin{aligned} \text{H. M.} &= \frac{N}{\sum_{i=1}^n f_i \left(\frac{1}{x_i} \right)} \\ &= \frac{26}{0.1648} \end{aligned}$$

$$\text{H.M} = 157.77$$

(c) H.M. for Continuous data:

$$\text{G. M.} = \text{Antilog } \frac{\sum_{i=1}^n f_i \log x_i}{N} \text{ with usual notations}$$

$$\text{The Harmonic mean H.M.} = \frac{N}{\sum_{i=1}^n f_i \left(\frac{1}{x_i} \right)}$$

Where x_i is the mid-point of the class interval

Example

Find the harmonic mean of the following distribution of data

Dividend yield (percent)	2 – 6	6 – 10	10 – 14
No. of companies	10	12	18

Solution:

Class Intervals	Mid-value (x_i)	No. of companies (f_i)	Reciprocal ($1/x_i$)	$f_i (1/x_i)$
2 – 6	4	10	$\frac{1}{4}$	2.5
6 – 10	8	12	$\frac{1}{8}$	1.5
10 – 14	12	18	$\frac{1}{12}$	1.5
Total		N = 40		5.5

$$\text{The harmonic mean is H.M.} = \frac{N}{\sum_{i=1}^n f_i \left(\frac{1}{x_i} \right)} = \frac{40}{5.5} = 7.27$$

Example: A's scooter gives an average of 40km a litre while B's scooter gives an average of 30km a litre . Find out the mean, if

- (i) each one of them travels 120 km.
- (ii) the petrol consumed by both of them is 2 litres per head.

Solution:

- (i) Here the distance is constant. Hence harmonic mean is appropriate.

$$\begin{aligned}
 HM &= \frac{n}{\frac{1}{a} + \frac{1}{b}} \\
 &= \frac{2}{\frac{1}{40} + \frac{1}{30}} = \frac{2}{\frac{7}{120}} \\
 &= \frac{2 \times 120}{7} = 34.3 \text{ km per litre}
 \end{aligned}$$

- (ii) Here the quantity of petrol consumed is fixed i.e 2 litres. Here the arithmetic mean will give the correct answer.

$$\begin{aligned}
 \bar{X} &= \frac{\text{Total distance Covered}}{\text{Total Petrol Consumed}} \\
 \bar{X} &= \frac{40 \times 2 + 30 \times 2}{4} = 35 \\
 \therefore \text{Average speed} &= 35 \text{ km per litre.}
 \end{aligned}$$

$\therefore$ Average speed = 35 km per litre.

Example : A person purchases tomatoes from each of the 4 places at the rate of 1kg., 2kg., 3kg., and 4kg. per rupee respectively .On the average, how many kilograms has he purchased per rupee?

Solution:

Since we are given rate per rupee, harmonic mean will give the correct answer.

$$\begin{aligned}
 HM &= \frac{n}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}} \\
 &= \frac{4}{\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}} \\
 &= \frac{4 \times 12}{25} \\
 &= 1.92 \text{ kg per rupee.}
 \end{aligned}$$

Merits of H.M:

- ☐ It is rigidly defined
- ☐ It is based on all the observations of the series
- ☐ It is suitable in case of series having wide dispersion
- ☐ It is suitable for further mathematical treatment
- ☐ It gives less weight to large items and more weight to small items

Limitations of H.M:

- ☐ It is difficult to calculate and is not understandable
- ☐ All the values must be available for computation
- ☐ It is not popular due to its complex calculation.
- ☐ It is usually a value which does not exist in series

When to use?

Harmonic mean is used to calculate the average value when the values are expressed as value/unit. Since the speed is expressed as km/hour, harmonic mean is used for the calculation of average speed.

Formulas:

$$\begin{aligned}\text{Arithmetic mean} &= \frac{x_1 + x_2 + \dots + x_n}{n} \\ \text{Geometric mean} &= \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n} \\ \text{Harmonic mean} &= \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}\end{aligned}$$

Ungroup data

$$A.M = \frac{\sum x_i}{n}$$

$$G.M = \text{Anti log} \left[\frac{1}{n} \sum \log x_i \right]$$

$$\frac{1}{N} \sum f_i \log x_i]$$

$$H.M = \frac{n}{\sum \frac{1}{x_i}}$$

Group data

$$A.M = \frac{\sum f_i x_i}{N}$$

$$G.M = \text{Anti log} [$$

$$H.M = \frac{N}{\sum \frac{f_i}{x_i}}$$

Where , n =total observations , $N=\sum f_i$,total frequency or total observations

The relationship among Arithmetic mean(A.M), Geometric mean(G.M) and Harmonic mean(H.M):

(1) If the numbers of the series are equal, then

A.M. = G.M. = H.M. For example 4,4,4

(2) If the numbers of the series are unequal

Then A.M.> G.M.> H.M. For example 4,7,9

(3) Thus, for non-zero positive data,

A.M. $\geq$ G.M. $\geq$ H.M.

(4) $GM = \sqrt{A.M. \times H.M.}$ or $(G.M.)^2 = A.M. \times H.M$

Relation between AM, GM & HM:

- The relation that associates the arithmetic mean, geometric mean and harmonic mean of a set of observations is given by:

$$AM \geq GM \geq HM$$

Proof: Let us consider two observations a and b. So, we have

$$AM = \frac{a+b}{2} \quad GM = \sqrt{a \cdot b} \quad HM = \frac{2ab}{a+b}$$

We, know that for any positive value of a and b $(\sqrt{a} - \sqrt{b})^2 > 0$

$$\Rightarrow a + b - 2\sqrt{a}\sqrt{b} \geq 0 \quad \Rightarrow a + b \geq 2\sqrt{a}\sqrt{b} \quad \Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$$

$$\Rightarrow AM \geq GM$$

Again, $a + b \geq 2\sqrt{a}\sqrt{b}$

$$\Rightarrow (a+b)\sqrt{ab} = 2(\sqrt{ab} \times \sqrt{ab})$$

$$\Rightarrow (a+b)\sqrt{ab} = 2ab \quad \Rightarrow \sqrt{ab} = \frac{2ab}{a+b}$$

$$\Rightarrow GM \geq HM$$

Thus, combining the two inequalities derived from above we have –

$$AM \geq GM \geq HM$$

Geometric mean(GM) equal to Geometric mean of Arithmetic mean(AM) and Harmonic mean (HM) or

$$GM = \sqrt{A.M. \times H.M} \quad \text{or} \quad (G.M.)^2 = A.M. \times H.M.$$

Let a & b two positive and non zero observations , then

$$AM = \frac{a+b}{2} \quad GM = \sqrt{a \cdot b} \quad HM = \frac{2}{\frac{1}{a} + \frac{1}{b}}$$

$$\text{So } GM = \sqrt{A.M. \times H.M} = \sqrt{\left(\frac{a+b}{2} \cdot \frac{2}{\frac{1}{a} + \frac{1}{b}}\right)} = \sqrt{\left(\frac{a+b}{2} \cdot \frac{2ab}{a+b}\right)} = \sqrt{ab} =$$

GM

Example: Compute the arithmetic mean (AM), geometric mean (GM) and harmonic mean (HM) of the following values: 8, 12, 14, 26, 5. and prove the relations.

Solution: We know, $AM(\bar{X}) = \frac{\sum X}{n} = \frac{8+12+14+26+5}{5} = \frac{65}{5} = 13$

$$G.M. = (x_1 \cdot x_2 \cdot x_3 \cdot x_4 \cdot x_5 \cdot x_6)^{\frac{1}{n}} = (8 \times 12 \times 14 \times 26 \times 5)^{\frac{1}{5}} = (174720)^{0.2} = 11.181$$

$$H.M = \frac{5}{\frac{1}{8} + \frac{1}{12} + \dots + \frac{1}{5}}$$

$$H.M = \frac{5}{0.478}$$

$$HM = 10.46$$

Then $A.M. > G.M. > H.M.$, $13 > 11.181 > 10.46$

$$\text{And } GM = \sqrt{A.M. \times H.M} = \sqrt{13 \cdot 10.46} = 11.66$$

Example Verify the relationship among AM, GM and HM for the following data

X	7	10	13	16	19	22	25	28
f	10	22	24	28	19	9	12	16

Solution:

X	f	Xf	$\log X$	$f \log X$	f / X
7	10	70	0.8451	8.4510	1.4286
10	22	220	1	22.0000	2.2000
13	24	312	1.1139	26.7346	1.8462
16	28	448	1.2041	33.7154	1.7500
19	19	361	1.2788	24.2963	1.0000
22	9	198	1.3424	12.0818	0.4091
25	12	300	1.3979	16.7753	0.4800
28	16	448	1.4472	23.1545	0.5714
	$\Sigma f = N = 140$	$\Sigma fX = 2357$		$\Sigma f \log x = 167.209$	$\Sigma \frac{f}{x} = 9.6852$

Table : 8.10

$$AM = \frac{\sum fX}{N} = \frac{2357}{140} = 16.84$$

$$GM = \text{Anti log} \left(\frac{\sum f \log X}{N} \right) = \text{Anti log} \left(\frac{167.209}{140} \right) = \text{Anti log}(1.1944) = 15.65$$

$$HM = \frac{N}{\Sigma \left(\frac{f}{X} \right)} = \frac{140}{9.6852} = 14.46$$

i.e. $16.84 > 15.65 > 14.46$

$\therefore AM > GM > HM$